

Unifying Special Relativity and Quantum Mechanics

Lecture 25 — Goldstone Bosons: Field Redefinitions, Effective Field Theory, and the Goldstone Theorem

(The Higgs and the Goldstone, non-linear field redefinitions, why the resulting non-renormalizable interactions are harmless, the EFT of a Goldstone boson and amplitude matching, and the general $G \rightarrow H$ construction)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–24: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, pencil). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination. This scan is the *direct continuation* of Lecture 24: that one ran [p. 1*****]–[p. 22*****] and this one picks up at [p. 23*****] and runs to [p. 42*****]. The numbering does not restart, so the twelve-star markers carry straight on. As with Lecture 24, the scan is clean: twenty sheets, circled numbers 23–42 consecutive, no duplicates, nothing missing, nothing out of order. PDF page n of this file is sheet $22 + n$.

The lecture ends mid-argument. Sheet [p. 42*****] differentiates the invariance condition of the potential, writes the resulting $\dim G$ equations, and then stops with the line “we evaluate this equation on the vacuum ϕ_0 ”. That evaluation *is* the proof of Goldstone’s theorem, and it evidently continues in the next lecture. Section 9 records the fragment exactly as it stands, and a blue box carries the argument through to the end so that you can see where it was going.

Two slips flagged in place.

1. In the yellow vertex box on [p. 30*****] the middle term is written $2(p_1 \cdot p_3)(2p_2 \cdot p_4)$; for the three terms to match the line above it must be $(2p_1 \cdot p_3)(2p_2 \cdot p_4)$ (Section 8).
2. On [p. 35*****] and [p. 42*****] the second-derivative symbol is written with a single ∂ in the numerator ($\partial V / \partial \phi^a \partial \phi^b$) where $\partial^2 V$ is meant. Written correctly throughout here; it is pure handwriting shorthand, not a mistake of substance.

A notation reminder carried over from Lecture 24. Two parametrizations of the same theory are in play throughout Part I: the *Cartesian* one with fields χ (massive) and θ (massless), and the *exponential* one with h (the Higgs) and π (the Goldstone). The notes call the corresponding Lagrangians $\mathcal{L}_\theta$ and $\mathcal{L}_\pi$. They describe the same physics; the whole of [p. 23*****]–[p. 27*****] is about the price and the reward of passing from one to the other. Also, f_π has now been set equal to v everywhere, so the exponent is written $e^{i\pi/v}$.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 24

Lecture 24 took two real scalars with a global $SO(2) \cong U(1)$ symmetry and a potential with $m^2 = -\mu^2 < 0$, $\lambda > 0$. The minimum of the potential is not a point but a circle $\phi_1^2 + \phi_2^2 = v^2$, $v^2 = \mu^2/\lambda$ — a continuum of degenerate vacua. The system picks one, $\langle \phi \rangle = (v, 0)^T$, and the ground state thereby fails to be invariant under the symmetry even though the Lagrangian still is.

Expanding around the chosen vacuum gave **one massive field**, with $m^2 = 2\mu^2$, and **one massless field**. Geometrically: climbing the wall of the trough costs energy; moving along

the trough costs nothing. Rewriting the theory in polar field variables, $\varrho = \frac{1}{\sqrt{2}}(v+h)e^{i\pi/v}$, made the reason transparent — the potential depends only on $\varrho^*\varrho = \frac{1}{2}(v+h)^2$, so π drops out of it entirely and survives only through $\partial_\mu\pi$. The residual symmetry is the shift $\pi \rightarrow \pi + c$, which is the original $U(1)$ realized *non-linearly*.

This lecture picks up exactly there and does three things. First it names the two particles (Higgs, Goldstone) and confronts an apparent problem: the exponential Lagrangian has non-renormalizable interactions, and Lectures 14 and 23 taught that these signal an incomplete theory. Sheets [p. 23*****]–[p. 27*****] show that this time they do not — a scattering amplitude is computed and the dangerous energy growth cancels. Second, [p. 28*****]–[p. 33*****] build the effective theory of the Goldstone alone, and match it onto the full theory to fix its coefficient. Third, [p. 34*****]–[p. 42*****] generalizes everything from $SO(2)$ to an arbitrary compact G , sets up the broken/unbroken split of the generators, and starts the proof of Goldstone’s theorem.

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PART I — Two descriptions of the broken phase [p. 23*****]— [p. 27*****]

1 The Higgs field and the Goldstone boson; the two Lagrangians [p. 23*****]

The field $h(x)$ (massive in the broken phase) is called the [Higgs field](#).

The field $\pi(x)$ (massless in the broken phase) is called a [Goldstone boson](#).

TECHNICAL COMMENT

We have two descriptions of the broken phase.

$$\mathcal{L}_\theta \equiv \frac{1}{2}(\partial_\mu \chi)(\partial^\mu \chi) - \frac{1}{2}(2\mu^2)\chi^2 + \frac{1}{2}(\partial_\mu \theta)(\partial^\mu \theta) - \frac{\lambda}{4}(\chi^4 + \theta^4 + 4v\chi^3 + 2\chi^2\theta^2 + 4v\chi\theta^2)$$

$$\mathcal{L}_\pi \equiv \frac{1}{2}(\partial_\mu h)(\partial^\mu h) - \frac{1}{2}(2\mu^2)h^2 + \frac{1}{2}(\partial_\mu \pi)(\partial^\mu \pi) + \frac{1}{2v^2}(h^2 + 2hv)(\partial_\mu \pi)(\partial^\mu \pi) - v\lambda h^3 - \frac{\lambda}{4}h^4$$

The two theories are related by a [change of variables in field space](#).

$$v + \chi(x) + i\theta(x) = \left(v + h(x)\right) e^{i\pi(x)/v}$$

⏟ let us expand the exponential

Added explanation (not in the original notes)

Where the two boxes come from. They are, verbatim, the two broken-phase Lagrangians derived in Lecture 24 — $\mathcal{L}_\theta$ on [p. 13*****] (Cartesian shift $\phi_1 = v + \chi$, $\phi_2 = \theta$) and $\mathcal{L}_\pi$ on [p. 19*****] (exponential parametrization $\varrho = \frac{1}{\sqrt{2}}(v + h)e^{i\pi/v}$). Nothing new is being introduced here; the two results are simply put side by side so that they can be compared.

Why the relation is that one. In the Cartesian variables $\varrho = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2) = \frac{1}{\sqrt{2}}(v + \chi + i\theta)$; in the exponential ones $\varrho = \frac{1}{\sqrt{2}}(v + h)e^{i\pi/v}$. Equating the two and cancelling the $1/\sqrt{2}$ gives exactly the yellow box. So this single equation contains the entire dictionary between the two descriptions.

The names. “Higgs” for the massive radial mode and “Goldstone” for the massless angular mode are the standard names, and they are being introduced here even though no gauge field is present — in this global model the Higgs is just a massive scalar and the Goldstone stays physical. The reason for the names is what happens when the symmetry is gauged (next lecture): the Goldstone is eaten, the radial mode survives as the Higgs boson.

2 Expanding the exponential: a non-linear field redefinition

[p. 24*****]

$$\begin{aligned}
 (v+h)\left(1+i\frac{\pi}{v}-\frac{1}{2}\left(\frac{\pi}{v}\right)^2+\cdots\right) &= \\
 = v+i\pi-\frac{\pi^2}{2v}+h+i\frac{h\pi}{v}+\cdots &= \\
 = \left(v+h-\frac{\pi^2}{2v}+\cdots\right)+i\left(\pi+\frac{\pi h}{v}+\cdots\right) &
 \end{aligned}$$

So that we can also write

$$\begin{aligned}
 \chi(x) &= h(x) - \frac{\pi(x)^2}{2v} + \cdots \\
 \theta(x) &= \pi(x) + \frac{\pi(x)h(x)}{v} + \cdots
 \end{aligned}$$

This is known as a “non-linear field redefinition”.

This transformation does change the Lagrangian ($\mathcal{L}_\theta \rightarrow \mathcal{L}_\pi$) but does not change the physics (in particular, on-shell S -matrix elements are the same).

Even though the physics does not change, there are field variables more suited than others $\rightsquigarrow$ in this case, the exponential parametrization with the Higgs and the Goldstone makes the physics of SSB more transparent.

$\hookrightarrow$ There is, however, a price to pay for making the Goldstone boson shift symmetry manifest: $\mathcal{L}_\pi$ has non-renormalizable interactions.

Added explanation (not in the original notes)

Read the two lines carefully — they are the point of the whole first half. A *linear* field redefinition ($\chi \rightarrow a\chi + b\theta$) is a change of basis and is entirely harmless. What happens here is *non-linear*: χ is expressed in terms of h and π with a quadratic term $-\pi^2/2v$, and θ with a term $\pi h/v$. Such redefinitions change the Lagrangian, change the Green’s functions, change the Feynman rules — and leave the S -matrix alone.

Why the S -matrix does not care (the “equivalence theorem”, Chisholm 1961, Kamefuchi–O’Raifeartaigh–Salam 1961). If $\varphi' = \varphi + \mathcal{O}(\varphi^2)$, then φ and φ' create the same one-particle states from the vacuum: the extra pieces are quadratic in the fields, so their matrix element between the vacuum and a one-particle state vanishes. Two fields differing by such terms are said to *interpolate* the same particle, and the LSZ reduction — which cares only about the pole of the propagator and the residue there — gives identical on-shell amplitudes. Off-shell quantities (Green’s functions, the effective potential away from its minimum, individual diagrams) do change, and there is no contradiction: they are not observables.

Why $\mathcal{L}_\pi$ has non-renormalizable terms, in one line. Look at $\frac{1}{2v^2}h^2(\partial\pi)(\partial\pi)$: four fields and two derivatives, so mass dimension $4+2=6$, and the coefficient $1/2v^2$ carries dimension -2 . In the classification of Lecture 13 that is an irrelevant, non-renormalizable operator. $\mathcal{L}_\theta$, by contrast, contains only dimension-4 operators. Same theory, different bookkeeping — and the next three sheets explain why that is not a disaster.

3 Non-renormalizable interactions of two different kinds

[p. 25*****]

This is a generic fact: if we start from a renormalizable Lagrangian (like $\mathcal{L}_\theta$) and perform a non-linear field redefinition, we end up with a new Lagrangian with non-renormalizable interactions (like $\mathcal{L}_\pi$).

↔ These non-ren. interactions, however, are not of the same sort compared to the one we discussed before.

Previously, we found non-renormalizable interactions that were the signal that the theory was not complete.

↔ A trademark for this fact are amplitudes that grow with energy.

To see the difference, we now compute a scattering amplitude with $\mathcal{L}_\pi$ and show that it does not grow with energy.

We consider the scattering process

$$\pi(p_1) + \pi(p_2) \longrightarrow \pi(p_3) + \pi(p_4)$$

at the tree level.

There is only one interaction relevant for this process.

Added explanation (not in the original notes)

The distinction being drawn is the single most important idea in this lecture. Both kinds of theory have operators of dimension > 4 with dimensionful couplings. They differ in whether the operators are *physically* there:

- *Genuine* non-renormalizable interactions — Fermi theory (G_F , Lecture 23), massive Yang–Mills ($g^2 E^2/M^2$, Lecture 23), gravity (E^2/M_P^2 , Lecture 21). Here amplitudes really do grow with E , unitarity really is violated at a definite scale, and new degrees of freedom really must appear there (the W , the Higgs, whatever completes gravity).
- *Spurious* ones — the case at hand. They are an artefact of the coordinates chosen in field space. The underlying theory is $\mathcal{L}_\theta$, manifestly renormalizable, so no amplitude can misbehave. The dangerous-looking vertices must therefore cancel among themselves in every on-shell amplitude, and [p. 26*****]–[p. 27*****] verify that they do.

So “non-renormalizable operators present” is not the diagnostic; “amplitude grows with energy” is. The first can be created or destroyed by a change of variables; the second cannot, because it is a statement about the S -matrix.

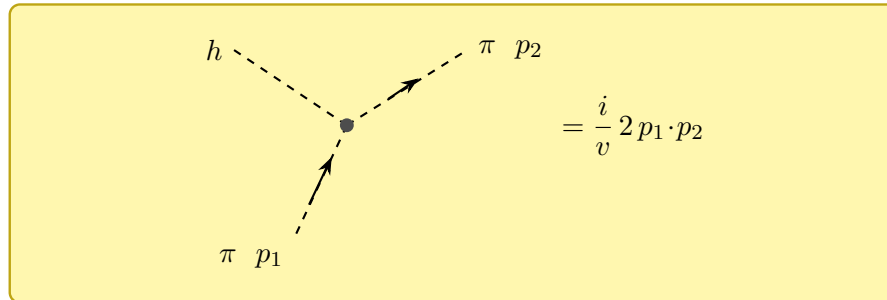
Why only one interaction matters at tree level. For $\pi\pi \rightarrow \pi\pi$ we need four external π legs. In $\mathcal{L}_\pi$ the candidates are $\frac{1}{v}h(\partial\pi)^2$ and $\frac{1}{2v^2}h^2(\partial\pi)^2$; the latter has two h ’s, so at tree level it would need two internal h propagators and cannot connect four external pions with a single vertex. Only $\frac{1}{v}h(\partial\pi)^2$ contributes — twice, with a Higgs exchanged between the two vertices. There is no direct π^4 vertex at all in $\mathcal{L}_\pi$: the shift symmetry forbids any non-derivative π coupling, and no four- π derivative operator was generated. That the whole amplitude proceeds through Higgs exchange is exactly why the answer will know about m_h .

4 The vertex, the three diagrams, and the s -channel amplitude

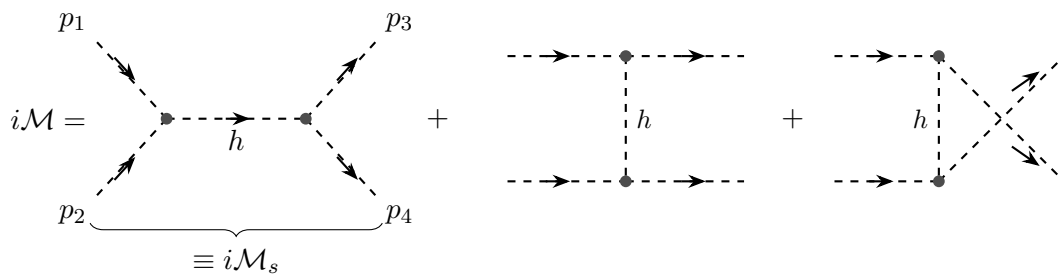
[p. 26 *****]

$$\mathcal{L}_{\text{int}} = \frac{1}{v} h (\partial_\mu \pi)(\partial^\mu \pi)$$

The Feynman rule is



The relevant diagrams are



The s -channel amplitude is

$$i\mathcal{M}_s = \frac{2i}{v} (-p_1 \cdot p_2) \frac{i}{s - m_h^2} \left(\frac{2i}{v} \right) (-p_3 \cdot p_4) = \left(\frac{-i}{v^2} \right) \frac{(2p_1 \cdot p_2)(2p_3 \cdot p_4)}{s - m_h^2}$$

Since Goldstones are massless, we have

$$\begin{cases} s = 2p_1 \cdot p_2 = 2p_3 \cdot p_4 \\ t = -2p_1 \cdot p_3 = -2p_2 \cdot p_4 \\ u = -2p_1 \cdot p_4 = -2p_2 \cdot p_3 \end{cases}$$

We thus rewrite

Added explanation (not in the original notes)

Deriving the vertex rule. From $\mathcal{L}_{\text{int}} = \frac{1}{v} h \partial_\mu \pi \partial^\mu \pi$, the vertex is $i \times \frac{1}{v} \times 2$ (the factor 2 from the two ways of assigning the two π fields to the two external pion lines) times the two momentum factors. With the conventions of the picture — p_1 *incoming* ($\partial_\mu \pi \rightarrow -ip_{1\mu}$) and p_2 *outgoing* ($\partial_\mu \pi \rightarrow +ip_{2\mu}$) — one gets $\frac{2i}{v} (-ip_1) \cdot (+ip_2) = \frac{i}{v} 2 p_1 \cdot p_2$, which is the boxed rule. When *both* pions are incoming, as at the left-hand vertex of the s -channel diagram, flip $p_2 \rightarrow -p_2$ and the rule becomes $\frac{2i}{v} (-p_1 \cdot p_2)$ — exactly what is used in the line above. This is why a sign appears there that is absent from the boxed rule; it is not an inconsistency.

Mandelstam variables for massless particles. With all $p_i^2 = 0$ and momentum conservation $p_1 + p_2 = p_3 + p_4$:

$$s = (p_1 + p_2)^2 = 2p_1 \cdot p_2, \quad t = (p_1 - p_3)^2 = -2p_1 \cdot p_3, \quad u = (p_1 - p_4)^2 = -2p_1 \cdot p_4,$$

and $s + t + u = \sum_i m_i^2 = 0$. **Keep that last identity in view** — it is the entire mechanism behind the cancellation on the next sheet.

Why the vertex carries two powers of momentum. Every π enters through $\partial_\mu \pi$, so every π leg brings a factor of its momentum. This is the shift symmetry at work in the Feynman rules, and it is why the amplitude is naïvely $\mathcal{O}(E^4/v^2 m_h^2)$ — four powers of momentum from two vertices, and a propagator. The whole question is whether that growth survives.

5 The full amplitude and its high-energy limit [p. 27*****]

$$i\mathcal{M}_s = \left(\frac{-i}{v^2}\right) \frac{s^2}{s - m_h^2}$$

The other amplitudes can be computed by substituting

$$s \rightarrow t \quad (\text{for } i\mathcal{M}_t)$$

$$s \rightarrow u \quad (\text{for } i\mathcal{M}_u)$$

The amplitude is

$$i\mathcal{M} = \frac{-i}{v^2} \left(\frac{s^2}{s - m_h^2} + \frac{t^2}{t - m_h^2} + \frac{u^2}{u - m_h^2} \right)$$

We now take the high energy limit $s, t, u \gg m_h^2$

$$i\mathcal{M} \stackrel{s, t, u \gg m_h^2}{=} -\frac{i}{v^2} (s + t + u) = 0$$

The amplitude does not grow with energy, we do not have any problem.

Non ren. interaction in this case arise not because our description is incomplete, but because we introduced non linear field redefinitions. In fact, they do not pose inconsistencies at high energies.

Added explanation (not in the original notes)

The cancellation, step by step. At large s, t, u expand each term:

$$\frac{s^2}{s - m_h^2} = s + m_h^2 + \frac{m_h^4}{s} + \dots$$

and similarly for t and u . Summing the three leading pieces gives $s + t + u = 0$ for massless external particles. The $\mathcal{O}(E^2)$ growth cancels identically, leaving a constant $3m_h^2/v^2$ plus terms falling as $1/E^2$. **No individual diagram is well behaved; only the sum is.** That is the signature of a spurious problem: had the theory really been incomplete, no conspiracy among the diagrams would have saved it.

Contrast with massive Yang–Mills (Lecture 23). There the analogous sum did *not* cancel, the residual growth was $g^2 E^2/M^2$, and unitarity failed at $E \sim M/g$. The difference is not technique but content: massive Yang–Mills was missing a degree of freedom, whereas $\mathcal{L}_\pi$ is a complete renormalizable theory wearing a disguise.

The cross-check that makes the result obvious. Compute the same process with $\mathcal{L}_\theta$ instead. There the interactions are the ordinary quartic ones $-\frac{\lambda}{4}(\theta^4 + 2\chi^2\theta^2 + 4v\chi\theta^2)$, every coupling is dimensionless, and no amplitude can grow with energy for trivial power-counting reasons. Since the two Lagrangians are related by a field redefinition, the on-shell amplitude must be the same — so the cancellation just verified was guaranteed in advance. The calculation is worth doing anyway, because in a case where only the non-linear variables are available (chiral perturbation theory, for instance) this is the only way to see it.

The low-energy limit is the interesting one. For $s, t, u \ll m_h^2$ each denominator becomes $-m_h^2$, and

$$i\mathcal{M} \simeq \frac{i}{v^2 m_h^2} (s^2 + t^2 + u^2),$$

which *does* grow with energy — as E^4 . That is not a contradiction: this expression is only valid below m_h , and above that scale the exact formula takes over and flattens out. This limit is precisely what the effective theory of the next section must reproduce, and the matching on [p. 32*****] is exactly this comparison.

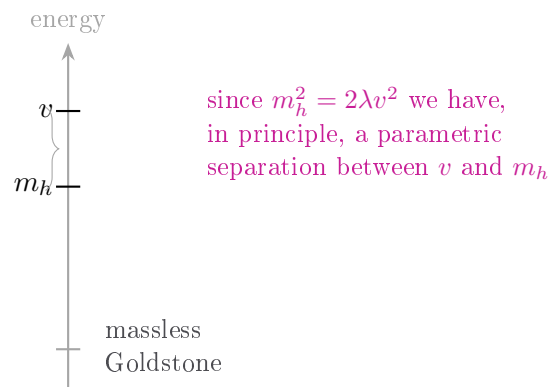
PART II — Effective field theory for Goldstone bosons [p. 28*****]— [p. 33*****]

6 The question: a Lagrangian for the Goldstone alone [p. 28*****]

■ EFT for Goldstone bosons

Consider our theory with SSB. In the broken phase, we have

Suppose that we are interested in the case of energies $E \ll m_h$. Is it possible to construct a Lagrangian that only describes the Goldstone?



What are the rules to construct such Lagrangian?

i) The Lagrangian we seek must contain $\pi(x)$ as dynamical d.o.f.

↪ At energies $E \ll m_h$ the Higgs mode is non-dynamical.

ii) The Goldstone boson is a real scalar field and, since it arises from the breaking of $SO(2)$, it enjoys a shift symmetry.

↪ This is the trademark that distinguishes $\pi(x)$ from being a generic scalar.

Added explanation (not in the original notes)

What “effective field theory” means here. The general recipe (the same one used for gravity in Lecture 21 and for Fermi theory in Lecture 23): decide which particles are light enough to be produced at the energies of interest, keep only those as fields, and write the most general Lagrangian built out of them consistent with the symmetries, organized in increasing mass dimension. Heavy states are not ignored — their effects survive in the coefficients of the higher-dimension operators, suppressed by powers of the heavy mass. Here the light field is π , the heavy state is h , and the coefficient c will turn out to encode m_h .

On the parametric separation. Since $m_h^2 = 2\mu^2 = 2\lambda v^2$, we have $m_h = \sqrt{2\lambda} v$. For $\lambda \ll 1$ the Higgs is much lighter than v , so there is a window $m_h \ll E \ll v$ — but the EFT of the Goldstone alone is valid *below* m_h . Note that v is the scale that sets the size of the Goldstone’s interactions, and m_h is the scale at which the EFT breaks down; they are genuinely different numbers, and the whole usefulness of the construction rests on that.

Rule (ii) is doing all the work. Without the shift symmetry, the most general Lagrangian for a real scalar would begin $\frac{1}{2}(\partial\pi)^2 - \frac{1}{2}m^2\pi^2 - \frac{g}{3!}\pi^3 - \frac{\lambda'}{4!}\pi^4$, all renormalizable and all symmetry-allowed. Imposing $\pi \rightarrow \pi + c$ kills every term without a derivative — mass, cubic, quartic — and leaves only $(\partial\pi)^2$ at the renormalizable level. The masslessness is not an input; it is a consequence.

7 The rules, and the leading operators [p. 29*****]

We construct a Lagrangian compatible with the following rules.

- At the renormalizable level, the only allowed term is

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \pi)(\partial^\mu \pi)$$

Free theory for a massless scalar field

- If we do include non-renormalizable interactions, we write

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \pi)(\partial^\mu \pi) + c \left((\partial_\mu \pi)(\partial^\mu \pi) \right)^2$$

dimension $[M]^8$

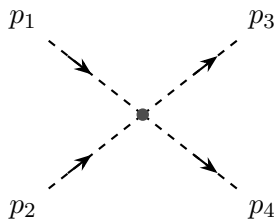
$$[c] = [M]^{-4}$$

Let us now compute the scattering of Goldstone bosons in this theory.

We consider the interaction

$$\mathcal{L}_{\text{int}} = c (\partial_\mu \pi)(\partial^\mu \pi)(\partial_\rho \pi)(\partial^\rho \pi)$$

This interaction gives rise to a vertex of the kind



Each field $\partial_\mu \pi \sim -ip_\mu a + ip_\mu a^\dagger$ has the capability of creating or destroying a scalar particle.

Added explanation (not in the original notes)

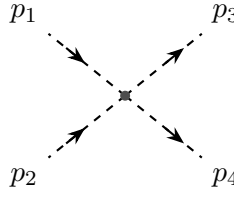
Why $((\partial\pi)^2)^2$ is the first interaction. Everything must be built from $\partial_\mu \pi$ (shift symmetry) and be Lorentz invariant. Contract indices in pairs: with two $\partial\pi$'s the only invariant is $(\partial_\mu \pi)(\partial^\mu \pi)$, which is the kinetic term. With four, the only Lorentz scalar with no additional derivatives is $((\partial_\mu \pi)(\partial^\mu \pi))^2$; the alternative contraction $(\partial_\mu \pi \partial_\nu \pi)(\partial^\mu \pi \partial^\nu \pi)$ is *the same thing*, since each $\partial_\mu \pi$ is a commuting object. Terms with $\partial^2 \pi$ can be removed using the leading equation of motion $\square \pi = 0$, or equivalently by a further field redefinition. Hence a single operator at this order.

The counting. $[\pi] = 1$, so $[\partial_\mu \pi] = 2$ and the operator has dimension 8. The Lagrangian density must have dimension 4, so $[c] = -4$: an inverse mass to the fourth. The matching on [p. 32*****] will identify that mass with m_h .

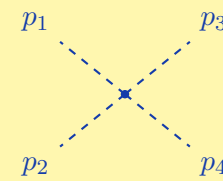
This construction is the model for chiral perturbation theory. Replace $SO(2) \rightarrow SU(2)_L \times SU(2)_R$ broken to $SU(2)_V$, and the Goldstones are the three pions; the analogue of $\frac{1}{2}(\partial\pi)^2 + c((\partial\pi)^2)^2$ is the leading chiral Lagrangian plus the L_1, L_2, L_3 terms of Gasser and Leutwyler, and low-energy $\pi\pi$ scattering data are used to measure exactly the coefficients that the matching below computes from a model. The notation $f_\pi, \pi(x)$ throughout these two lectures is a deliberate signal of that.

8 The four-pion vertex and its Feynman rule [p. 30*****]

destroying a scalar particle. The Feynman rule is



$$\begin{aligned}
 &= (-1)(-i)c \left[\underbrace{(-ip_{1\mu})(-ip_2^\mu)}_{(\partial_\mu \pi)(\partial^\mu \pi)} \underbrace{(ip_{3\rho})(ip_4^\rho)}_{(\partial_\rho \pi)(\partial^\rho \pi)} \times 8 \right. \\
 &\quad + (-ip_{1\mu})(ip_3^\mu)(-ip_{2\rho})(ip_4^\rho) \times 8 \\
 &\quad \left. + (-ip_{1\mu})(ip_4^\mu)(-ip_{2\rho})(ip_3^\rho) \times 8 \right] = \\
 &= 8ic \left[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_3)(p_2 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) \right]
 \end{aligned}$$



$$\begin{aligned}
 &\equiv 2ic \left[(2p_1 \cdot p_2)(2p_3 \cdot p_4) + (2p_1 \cdot p_3)(2p_2 \cdot p_4) \right. \\
 &\quad \left. + (2p_1 \cdot p_4)(2p_2 \cdot p_3) \right]
 \end{aligned}$$

The scattering amplitude for the process $\pi(p_1) + \pi(p_2) \rightarrow \pi(p_3) + \pi(p_4)$ is given by

$$i\mathcal{M} = 2ic(s^2 + t^2 + u^2)$$

The amplitude grows with energy; this means that the Lagrangian

Added explanation (not in the original notes)

Slip flagged. In the yellow box the middle term is written $2(p_1 \cdot p_3)(2p_2 \cdot p_4)$; the factors of 2 must be distributed as $(2p_1 \cdot p_3)(2p_2 \cdot p_4)$ for the three terms to be the same structure and to reproduce the $8ic[\dots]$ line above. Corrected here.

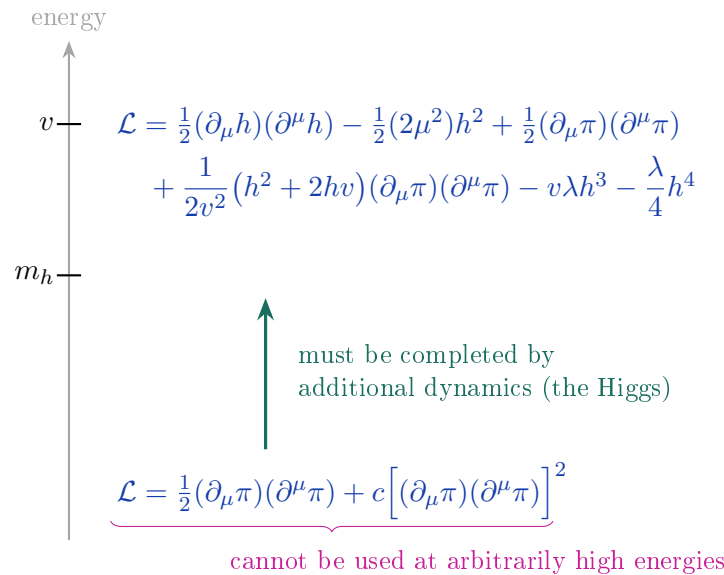
Where the numbers come from. The vertex is $i\mathcal{L}_{\text{int}}$ with all contractions summed. The operator has four π 's, and there are $4! = 24$ ways to attach them to the four external lines; these fall into 3 distinct pairings ($\{12\}\{34\}$, $\{13\}\{24\}$, $\{14\}\{23\}$), each occurring 8 times — hence the explicit $\times 8$ in each line. The overall (-1) in the notes is the usual $\mathcal{L} \rightarrow -\mathcal{L}$ sign bookkeeping; combined with the i 's from the four derivatives ($i^2 \times (-i)^2 = +1$ per pairing, up to the sign of the dot products) it gives the $+8ic$ shown.

From the vertex to $s^2 + t^2 + u^2$. Using the massless relations of [p. 26*****]: $(2p_1 \cdot p_2)(2p_3 \cdot p_4) = s \cdot s = s^2$, $(2p_1 \cdot p_3)(2p_2 \cdot p_4) = (-t)(-t) = t^2$, and likewise the third gives u^2 . Hence $i\mathcal{M} = 2ic(s^2 + t^2 + u^2)$ — note the contrast with [p. 27*****]: there the $\mathcal{O}(E^2)$ pieces cancelled because they combined into $s + t + u = 0$; here the $\mathcal{O}(E^4)$ pieces are *squares*, all of the same sign, and $s^2 + t^2 + u^2$ is strictly positive. Nothing can cancel.

9 The EFT must be completed; amplitude matching [p. 31*****]— [p. 32*****]

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \pi)(\partial^\mu \pi) + c \left((\partial_\mu \pi)(\partial^\mu \pi) \right)^2$$

only provides an EFT description of the Goldstone dynamics that must be completed at high energies.



It is possible to find the value of “ c ” that corresponds to the UV-completion given by the Higgs mode by performing the “[amplitude matching](#)”.

1) We computed the amplitude for the process $\pi\pi \rightarrow \pi\pi$ in the full theory. We find

$$i\mathcal{M} = \frac{(-i)}{v^2} \left(\frac{s^2}{s - m_h^2} + \frac{t^2}{t - m_h^2} + \frac{u^2}{u - m_h^2} \right)$$

2) We take the low energy limit $s, t, u \ll m_h^2$

$$i\mathcal{M} \simeq \frac{i}{v^2 m_h^2} (s^2 + t^2 + u^2)$$

3) We compare with the amplitude of the EFT

$$i\mathcal{M} \simeq \frac{i}{v^2 m_h^2} (s^2 + t^2 + u^2)$$

amplitude in the low energy limit of
the UV-complete model

$\stackrel{!}{=}$

$$i\mathcal{M} = 2ic(s^2 + t^2 + u^2)$$

amplitude in the EFT

We find $2c = \frac{1}{v^2 m_h^2} \xrightarrow{\text{using}} \frac{1}{m_h^2} = 2\lambda v^2$

$$c = \frac{\lambda}{m_h^4}$$

The $\pi\pi$ -scattering amplitude computed in the EFT grows with energy with the scaling

$$i\mathcal{M} = \mathcal{O}\left(\frac{\lambda E^4}{m_h^4}\right) \rightsquigarrow \text{cutoff at } E \sim \frac{m_h}{\lambda^{1/4}}$$

Added explanation (not in the original notes)

The logic of matching, in one paragraph. Two descriptions of the same physics overlap in a window of energies where both are valid — here E well below m_h , where the full theory can be Taylor-expanded and the EFT has not yet broken down. Demand that they give the same answer for a physical quantity in that window, and the unknown coefficients of the EFT are determined in terms of the parameters of the full theory. This is the same manoeuvre by which G_F was matched onto g^2/M_W^2 in Lecture 23. The EFT does not *predict* c ; it predicts the *form* $s^2 + t^2 + u^2$, and c is an input to be either measured or matched.

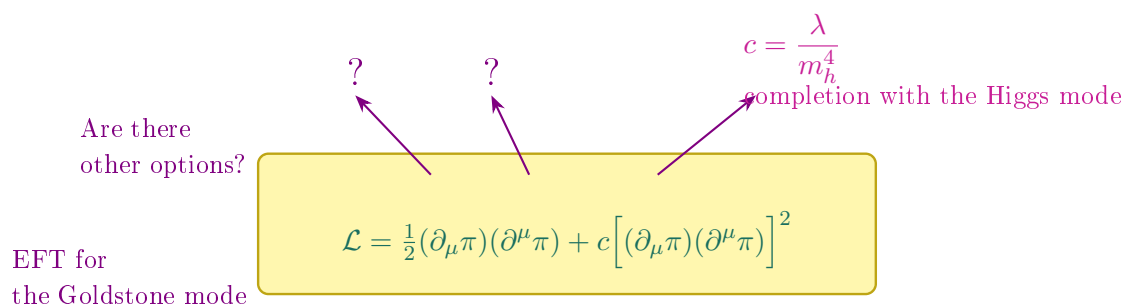
Doing step 2 explicitly. For $|s| \ll m_h^2$, $\frac{s^2}{s - m_h^2} = -\frac{s^2}{m_h^2} \left(1 + \frac{s}{m_h^2} + \dots\right)$, so the leading term of $i\mathcal{M}$ is $\frac{-i}{v^2} \cdot \left(-\frac{s^2 + t^2 + u^2}{m_h^2}\right) = \frac{i}{v^2 m_h^2} (s^2 + t^2 + u^2)$. Note that the leading behaviour is now E^4 , not the E^2 that cancelled: expanding the propagator has traded the exact cancellation for a suppressed remainder, which is exactly what integrating out a heavy particle always does.

Two ways to read c . From $2c = 1/(v^2 m_h^2)$ one gets, using $m_h^2 = 2\lambda v^2$, $c = \lambda/m_h^4$ — or, equivalently, $c = 1/(4\lambda v^4)$. The first form makes the cutoff transparent: $i\mathcal{M} \sim cE^4 = \lambda E^4/m_h^4$, which reaches $\mathcal{O}(1)$ at $E \sim m_h/\lambda^{1/4}$. For weak coupling $\lambda \ll 1$ that is somewhat *above* m_h , which is consistent: the EFT starts failing where the Higgs pole is approached, and fails completely by the time unitarity is saturated.

The two energy-growth statements are not in conflict. [p. 27*****] showed that the amplitude of the *full* theory does not grow. [p. 30*****] showed that the amplitude of the *EFT* does. Both are right, and the difference is precisely the point of Part II: the full theory is complete, the EFT is a truncation valid only below m_h , and the growth of the truncated amplitude is the EFT announcing its own domain of validity.

10 Is the Higgs the only UV completion? [p. 33*****]–[p. 34*****]

Comment: Is the Higgs the only UV-completion of the Goldstone mode?



↔ Short answer: in principle yes. The point is that each UV-completion gives testable predictions.

Example:

- i) Suppose that we measure at low energy the scattering $\pi\pi \rightarrow \pi\pi$, so that we have access to a measurement of “ c ”.

- ii) The SSB predicts the existence of a scalar, the Higgs. Suppose that we found it, and measure its mass.
- iii) From m_h and $c = \lambda/m_h^4$ we extract λ and, therefore, v .
- iv) The model predicts that h will have self-interactions with strength controlled by λv and λ , completely fixed.

If we observe such self-interactions and find values consistent with the theoretical prediction, we test the model.

Added explanation (not in the original notes)

Why this is not a rhetorical question. The EFT $\frac{1}{2}(\partial\pi)^2 + c[(\partial\pi)^2]^2$ is agnostic about what sits above the cutoff. A heavy scalar (the linear sigma model of these two lectures) is one option; a heavy vector resonance, a composite sector, or a tower of states would all reproduce the same leading operator with a different value of c and, crucially, *different subleading operators*. The number c alone does not identify the completion — but the pattern of *several* coefficients, plus the spectrum, does.

Counting the parameters. The model here has two: λ and v (equivalently λ and m_h). Three low-energy observables — c , m_h , and the Higgs cubic self-coupling λv — are therefore *not* independent. Measure any two, predict the third. That is the sense in which the model is falsifiable, and it is exactly the logic of the Higgs self-coupling measurements now being attempted at the LHC: in the Standard Model λ_{hhh} is fixed once m_h and v are known, so any deviation signals a different UV completion.

The historical case. This entire chain of reasoning was run in reverse for the strong interactions. Low-energy $\pi\pi$ scattering was measured; the coefficients of the chiral Lagrangian were extracted; and the question “what completes it?” had the answer QCD — not a single heavy scalar but a whole confining gauge theory. The lesson is exactly the one in the magenta line: the EFT is the reliable part, the completion is a hypothesis to be tested.

PART III — The Goldstone theorem: the general construction

[p. 34*****]–[p. 42*****]

11 General setup: a scalar theory with a compact group G

[p. 34*****]

■ The Goldstone theorem

We try to set a more general discussion. We consider a scalar theory with a global symmetry group G compact, infinitesimally:

$$\phi^a(x) \xrightarrow{G} \phi'^a(x) = \phi^a(x) - i \alpha_A (T^A)^a_b \phi^b(x)$$

infinitesimal symmetry transformation

↑ multiplet of scalar fields

▷ $A = 1, \dots, \dim G$

$$[T^A, T^B] = i f_{ABC} T^C$$

The Lagrangian takes the form

$$\mathcal{L} = \mathcal{L}_{\text{KIN}} - V(\phi)$$

We have $\delta\mathcal{L} = 0$ if we transform the fields, by definition of internal symmetry.

↑ ↑
pure kinetic potential (including
Lagrangian mass term)

The condition $\delta\mathcal{L} = 0$ is verified independently by $\mathcal{L}_{\text{KIN}}$ and $V(\phi)$.

Added explanation (not in the original notes)

Why the two pieces must be separately invariant. $\mathcal{L}_{\text{KIN}}$ contains two derivatives and V contains none. A symmetry transformation with constant α_A does not change the number of derivatives on any term, so terms with different derivative counts cannot compensate each other. Hence $\delta\mathcal{L}_{\text{KIN}} = 0$ and $\delta V = 0$ separately. The rest of the argument uses only the second condition — the whole of Goldstone’s theorem is a statement about V .

Reading the transformation law. The fields sit in some real representation of G labelled by a ; the generators $(T^A)^a_b$ are the matrices of that representation, and the structure constants f_{ABC} are properties of the group, not of the representation. For G compact one can always choose a basis in which the T^A are hermitian and, for a real multiplet, purely imaginary and antisymmetric — which is why the explicit i appears in the transformation and why ϕ^a stays real. The $SO(2)$ example of Lecture 24 is the case $\dim G = 1$, $a = 1, 2$, $T = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$.

What the next eight sheets do. They repeat, in this general language, exactly the three steps of Lecture 24: find the minimum of V ; expand around it; read off the masses. The new ingredient is that “the mass” is now a *matrix*, and the theorem will be a statement about how many of its eigenvalues are forced to vanish.

12 Invariance of the potential; the minimum; shifted fields

[p. 35 *****]

From $\delta V = 0$, we get

$$\delta V = \frac{\partial V}{\partial \phi^a} \delta \phi^a = \frac{\partial V}{\partial \phi^a} (-i) \alpha_A (T^A)^a_b \phi^b \stackrel{!}{=} 0$$

$\rightsquigarrow$

$$\frac{\partial V}{\partial \phi^a} \alpha_A (T^A)^a_b \phi^b = 0$$

Invariance of the potential under the action of G .

We now minimize the energy. The kinetic part is minimized by considering field configurations that are constant in time and homogeneous in space.

As far as the potential is concerned, we suppose that it is minimized by some field configuration ϕ_0 .

$$\frac{\partial V}{\partial \phi^a} (\phi = \phi_0) = 0$$

$$\frac{\partial^2 V}{\partial \phi^a \partial \phi^b} (\phi = \phi_0) \text{ positive semi-definite}$$

(eigenvalues ≥ 0)

We introduce the shifted fields

$$\phi^a(x) = \phi_0^a + \chi^a(x)$$

and expand the potential around the minimum. We write

Added explanation (not in the original notes)

The key equation is the grey box. Note that it holds for *all* ϕ , not just at the minimum — it is the statement that V is constant along every orbit of G in field space. And since the α_A are arbitrary and independent, it is really $\dim G$ separate equations, one per generator:

$$\frac{\partial V}{\partial \phi^a} (T^A)^a_b \phi^b = 0, \quad A = 1, \dots, \dim G.$$

Sheet [p. 42 *****] differentiates this identity once more and evaluates at ϕ_0 ; that single manoeuvre *is* Goldstone's theorem.

Why positive semi-definite and not positive definite. At a minimum the Hessian cannot have negative eigenvalues, but it may have zero ones — flat directions. In the Wigner–Weyl case there are none and all eigenvalues are strictly positive; in the spontaneously broken case there are exactly $\dim G - \dim H$ zeros. The notes' careful choice of words here already anticipates the result.

Note the reuse of the letter χ . In Lecture 24, χ was the specific massive Cartesian field of the $SO(2)$ model. Here χ^a is the general multiplet of fluctuations about ϕ_0 — some of its components will be massive, some massless. Same symbol, wider meaning.

13 The mass matrix [p. 36*****]

$$V(\phi) = V(\phi_0 + \chi) = V(\phi_0) + \cancel{\frac{\partial V}{\partial \phi^a}(\phi_0) \chi^a} + \frac{1}{2} \frac{\partial^2 V}{\partial \phi^a \partial \phi^b}(\phi_0) \chi^a \chi^b + \dots$$

(= 0)

$$V(\phi) = V(\phi_0) + \frac{1}{2} \frac{\partial^2 V}{\partial \phi^a \partial \phi^b}(\phi_0) \chi^a \chi^b + \dots$$

↔ This object is called “mass matrix”.
The eigenvalues of this matrix give the “mass spectrum” of the theory,

↑ Since we are interested in the mass spectrum of the theory, we only expand up to quadratic order.

$$M_{ab}^2 \equiv \frac{\partial^2 V}{\partial \phi^a \partial \phi^b}(\phi_0)$$

Before proceeding, let us discuss few examples.

Added explanation (not in the original notes)

Why the eigenvalues, and not the entries, are the masses. The quadratic Lagrangian is $\frac{1}{2}(\partial\chi^a)^2 - \frac{1}{2}M_{ab}^2\chi^a\chi^b$. The kinetic term is proportional to δ_{ab} , so it is invariant under orthogonal rotations of the χ^a ; M^2 is real symmetric, hence diagonalizable by such a rotation. In the rotated basis the fields decouple and each has a definite mass² equal to an eigenvalue. Fields in the original basis are generally *not* mass eigenstates — Example 3 below is precisely a case where they are not.

The counting to keep in mind. M^2 is a $(\dim \text{rep}) \times (\dim \text{rep})$ matrix. Goldstone’s theorem will say: the number of zero eigenvalues is at least $\dim G - \dim H$, where H is the subgroup leaving ϕ_0 invariant. Everything on [p. 39*****]–[p. 42*****] is the construction of that statement.

14 Three examples [p. 36*****]–[p. 38*****]

1) $G = SO(2)$ realized à la Wigner–Weyl

$$V = \frac{m^2}{2}(\phi_1^2 + \phi_2^2) + \frac{\lambda}{4}(\phi_1^2 + \phi_2^2)^2; \quad m^2, \lambda > 0$$

The vacuum state is $\phi_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

The mass matrix is given by

$$\frac{\partial^2 V}{\partial \phi_1^2} = m^2 + \lambda(3\phi_1^2 + \phi_2^2); \quad \frac{\partial^2 V}{\partial \phi_2^2} = m^2 + \lambda(\phi_1^2 + 3\phi_2^2)$$

$$\frac{\partial^2 V}{\partial \phi_1 \partial \phi_2} = 2\lambda \phi_1 \phi_2$$

$$\rightsquigarrow M_{ab}^2 = \begin{pmatrix} m^2 & 0 \\ 0 & m^2 \end{pmatrix}$$

2) $G = SO(2)$, SSB with $\phi_0 = (v, 0)^T$

$$V = -\frac{\mu^2}{2}(\phi_1^2 + \phi_2^2) + \frac{\lambda}{4}(\phi_1^2 + \phi_2^2)^2, \quad \phi_0 = \begin{pmatrix} v \\ 0 \end{pmatrix}$$

The mass matrix is given by

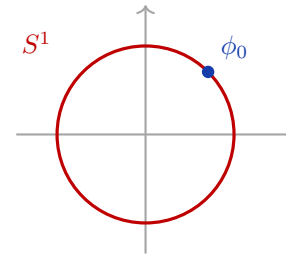
$$M_{ab}^2 = \begin{pmatrix} -\mu^2 + 3\lambda v^2 & 0 \\ 0 & -\mu^2 + \lambda v^2 \end{pmatrix} = \begin{pmatrix} 2\mu^2 & 0 \\ 0 & 0 \end{pmatrix}$$

↑ using $v^2 = \mu^2/\lambda$

3) $G = SO(2)$, SSB with $\phi_0 = \frac{1}{\sqrt{2}}(v, v)^T$

Different choice of vacuum which should give the same physics.

The mass matrix is given by



$$M_{ab}^2 = \begin{pmatrix} 2v^2\lambda - \mu^2 & v^2\lambda \\ v^2\lambda & 2v^2\lambda - \mu^2 \end{pmatrix} = \begin{pmatrix} \mu^2 & \mu^2 \\ \mu^2 & \mu^2 \end{pmatrix}$$

↔ The mass matrix is non-diagonal, but the eigenvalues are still $2\mu^2$ and 0, so the physical mass spectrum is the same.

The fact that the mass matrix is non-diagonal reflects the fact that, at the Lagrangian level, if we take the vacuum

$$\phi_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} v \\ v \end{pmatrix}$$

and introduce shifted fields

$$\phi = \begin{pmatrix} \frac{v}{\sqrt{2}} + \chi_1(x) \\ \frac{v}{\sqrt{2}} + \chi_2(x) \end{pmatrix}$$

we find, in the broken phase,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \chi_1)(\partial^\mu \chi_1) + \frac{1}{2}(\partial_\mu \chi_2)(\partial^\mu \chi_2) - \frac{\mu^2}{2}(\chi_1^2 + \chi_2^2 + 2\chi_1\chi_2) + \dots$$

Because of this “mixing” term, we cannot read the mass spectrum from this Lagrangian (χ_1 and χ_2 are not “mass eigenstates”) but we need to diagonalize the mass matrix.

Added explanation (not in the original notes)

Checking Example 2. With $\phi_0 = (v, 0)$: $\partial^2 V / \partial \phi_1^2 = -\mu^2 + \lambda(3v^2) = -\mu^2 + 3\mu^2 = 2\mu^2$; $\partial^2 V / \partial \phi_2^2 = -\mu^2 + \lambda v^2 = -\mu^2 + \mu^2 = 0$; and the off-diagonal entry $2\lambda\phi_1\phi_2$ vanishes since $\phi_2 = 0$. This reproduces exactly the result of Lecture 24 [p. 13*****]: the radial fluctuation χ has $m^2 = 2\mu^2$, the angular one θ is massless.

Checking Example 3. With $\phi_1^2 = \phi_2^2 = v^2/2$: $\partial^2 V / \partial \phi_1^2 = -\mu^2 + \lambda(3\frac{v^2}{2} + \frac{v^2}{2}) = -\mu^2 + 2\lambda v^2 =$

μ^2 , same for ϕ_2 ; and the off-diagonal entry is $2\lambda \cdot \frac{v^2}{2} = \lambda v^2 = \mu^2$. So $M^2 = \mu^2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, whose eigenvalues are $\mu^2(1 \pm 1) = \{2\mu^2, 0\}$ with eigenvectors $\frac{1}{\sqrt{2}}(1, 1)$ and $\frac{1}{\sqrt{2}}(1, -1)$. The massive direction points *along* ϕ_0 (radial) and the massless one perpendicular to it (tangent to the circle) — exactly as it must be.

The moral of Examples 2 and 3. The vacuum is a choice of coordinates; the mass matrix depends on it, the *spectrum* does not. Choosing ϕ_0 along an axis makes M^2 diagonal and the calculation short; any other choice mixes the fields and requires a diagonalization at the end. Nothing physical distinguishes them — which is just the statement, from Lecture 24 [p. 8*****], that all points of the vacuum manifold are equivalent.

Note what Example 1 shows. There $M^2 \propto \mathbb{K}$: the degeneracy is exact and enforced by the symmetry, because the unbroken G acts on the fields and commutes with M^2 , so mass eigenvalues come in complete multiplets of G . This is the general version of the “mass degeneracy is a trademark of Wigner–Weyl” statement from Lecture 24 [p. 7*****].

15 Broken and unbroken generators [p. 39*****]–[p. 40*****]

diagonalize the mass matrix.

We now ask what is the action of the symmetry group on the vacuum:

■ When the symmetry is realized à la Wigner–Weyl, the vacuum is left invariant by the action of G . Equivalently, the group generators annihilate the vacuum, $T^A \phi_0 = 0$.

■ In the case of SSB, the vacuum is not left invariant by the action of the group. Equivalently, the group generators do not annihilate the vacuum, $T^A \phi_0 \neq 0$.

The most general situation is known as “partial symmetry breaking”: some generators do annihilate the vacuum, and some generators do not annihilate the vacuum.

We introduce the following notation. We separate the generators T^A , $A = 1, \dots, \dim G$ of G into two categories.

$$\{T^A\} = \{T^a, X^\alpha\}$$

Generators that annihilate the vacuum

$$T^a \phi_0 = 0$$

a.k.a. “unbroken generators”

Generators that do not annihilate the vacuum

$$X^\alpha \phi_0 \neq 0$$

a.k.a. “broken generators”

We have the following results:

1) The unbroken generators T^a close a subalgebra.

↪ Consequently, $\exp(-i\alpha_a T^a)$ form a subgroup of G , which we denote as “ H ”.

Let us compute the commutator

$$[T^a, T^b] = if_{abC} T^C = if_{abc} T^c + if_{ab\alpha} X^\alpha$$

↪ we split the sum into broken and unbroken generators

⇓ we now apply this equation on the vacuum

$$[T^a, T^b] \phi_0 = if_{abc} T^c \phi_0 + if_{ab\alpha} X^\alpha \phi_0 ; \quad \text{we observe that}$$

Added explanation (not in the original notes)

Why “ $T^A\phi_0 = 0$ ” is the right way to say “the vacuum is invariant”. A finite transformation acts as $\phi_0 \rightarrow e^{-i\alpha_A T^A} \phi_0$. Demanding this equal ϕ_0 for all α is, infinitesimally, $T^A\phi_0 = 0$. So the two bullets are two ways of saying the same thing, and the second is the one that can be fed into algebra.

The three regimes. All generators unbroken ($H = G$): Wigner–Weyl, unique invariant vacuum, degenerate multiplets — Example 1. All broken ($H = \{e\}$): complete breaking — the $SO(2)$ model of Lecture 24, where the single generator does not annihilate $(v, 0)^T$. Some of each: partial breaking, $G \rightarrow H$. The Standard Model electroweak sector is the third case: $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$, with $\dim G = 4$, $\dim H = 1$, so three broken generators and three would-be Goldstones — which become the longitudinal modes of $W^\pm$ and Z .

Notation to keep straight. Latin a, b, c now label *unbroken* generators, Greek α, β label *broken* ones, and capital A, B, C run over all of them. Unfortunately a, b were also used as field-multiplet indices on [p. 34*****]–[p. 36*****]; from [p. 39*****] onwards they are generator labels. Context disambiguates, but it is worth noticing.

16 $G \rightarrow H$: the symmetry breaking pattern [p. 41***]**

$$\underbrace{[T^a, T^b]\phi_0}_{\text{by def.}=0} = \underbrace{if_{abc}T^c\phi_0}_{\text{by def.}=0} + if_{ab\alpha}X^\alpha\phi_0 \rightsquigarrow if_{ab\alpha}X^\alpha\phi_0 \stackrel{!}{=} 0$$

We find $if_{ab\alpha}X^\alpha\phi_0 = 0$; however, $X^\alpha\phi_0 \neq 0$ by definition of broken generators.

Consequently,

$$f_{ab\alpha} = 0, \quad \forall a, b, \alpha$$

Back to the commutator, we thus get

$$[T^a, T^b] = if_{abc}T^c$$

subalgebra $\mathfrak{h}$ of $\mathfrak{g}$

In these cases, we say that the symmetry G is spontaneously broken down to H .

$$G \longrightarrow H$$

“symmetry breaking pattern”

2) The subgroup H is not normal.

3) The broken generators do not form a subalgebra.

We can thus update our classification of generators, and write

Added explanation (not in the original notes)

The argument in one sentence. If two unbroken generators had a commutator with a component along a broken generator, that component would have to annihilate the vacuum — which broken generators by definition do not. Hence the commutator of two unbroken

generators is purely unbroken, and the T^a close.

Point 3, spelled out. Since $f_{ab\alpha} = 0$ and the structure constants of a compact group are totally antisymmetric, $f_{\alpha ab} = 0$ too — but $f_{\alpha\beta c}$ need not vanish. So

$$[T^a, T^b] \sim T^c, \quad [T^a, X^\alpha] \sim X^\beta, \quad [X^\alpha, X^\beta] \sim T^c + X^\gamma.$$

The middle relation says the broken generators transform in a *representation* of H — which is exactly why the Goldstone bosons form a multiplet of the unbroken group. The last says the broken generators do *not* close: they span the coset $\mathfrak{g}/\mathfrak{h}$, not a subalgebra. (When $[X, X] \sim T$ only, with no X on the right, the coset is called *symmetric*; that is the case in chiral symmetry breaking, and it is what makes the exponential parametrization work so cleanly.)

Point 2, “ H is not normal”. A normal subgroup would satisfy $gHg^{-1} = H$ for all $g \in G$, i.e. $[X^\alpha, T^a]$ would have to lie inside $\mathfrak{h}$. It does not — by the middle relation above it lies in the broken directions. Geometrically this is why the vacuum manifold G/H is a coset *space* and not a group: rotating the chosen vacuum by a broken generator lands on a different vacuum, whose stabilizer is a different (conjugate) copy of H .

Recover Lecture 24 as a special case. $G = SO(2)$, $\dim G = 1$, and the single generator is broken, so H is trivial, $\dim H = 0$, and there are no unbroken generators for the argument to constrain. The vacuum manifold is $G/H = SO(2) \simeq S^1$, the circle — exactly the picture on [p. 8*****].

17 The updated classification, and the start of the proof

[p. 42*****]

$$\{T^A\} = \left\{ \underbrace{T^1, \dots, T^{\dim H}}; X^\alpha \right\}$$

↓ form a subalgebra of dimension $\dim H$

▷ broken generators: their number is $\dim G - \dim H$

With this understanding in mind, let's go back to the equation (pg. [p. 35*****])

$$\frac{\partial V}{\partial \phi^a} \alpha_A (T^A)^a{}_b \phi^b = 0$$

Take derivative w.r.t. ϕ^c

$$\begin{aligned} \frac{\partial^2 V}{\partial \phi^a \partial \phi^c} \alpha_A (T^A)^a{}_b \phi^b + \frac{\partial V}{\partial \phi^a} \alpha_A (T^A)^a{}_b \delta_{bc} &= \\ = \alpha_A \left[\frac{\partial^2 V}{\partial \phi^a \partial \phi^c} (T^A)^a{}_b \phi^b + \frac{\partial V}{\partial \phi^a} (T^A)^a{}_c \right] &= 0 \end{aligned}$$

↓ Because of lin. independence, we write $\dim G$ equations

$$\frac{\partial^2 V}{\partial \phi^a \partial \phi^c} (T^A)^a{}_b \phi^b + \frac{\partial V}{\partial \phi^a} (T^A)^a{}_c = 0$$

We evaluate this equation on the vacuum ϕ_0

[the scan ends here]

Added explanation (not in the original notes) — Completing the argument — Goldstone's theorem

The scan stops one line short of the conclusion. Here it is.

Step 1. Evaluate the last equation at $\phi = \phi_0$. The second term contains $\partial V / \partial \phi^a(\phi_0)$, which vanishes because ϕ_0 is a stationary point. What survives is

$$M_{ac}^2 (T^A)^a{}_b \phi_0^b = 0 \quad \text{for every } A = 1, \dots, \dim G,$$

where $M_{ac}^2 = \partial^2 V / \partial \phi^a \partial \phi^c(\phi_0)$ is the mass matrix of [p. 36*****]. This is the whole theorem in one line: *for each generator, the vector $T^A \phi_0$ is a null eigenvector of the mass matrix.*

Step 2 — read off the two cases.

- If T^A is *unbroken*, $T^A \phi_0 = 0$ and the equation is $0 = 0$. No information.
- If $T^A = X^\alpha$ is *broken*, then $v^\alpha \equiv X^\alpha \phi_0 \neq 0$ is a genuine non-zero vector in field space, and $M^2 v^\alpha = 0$. So M^2 has a zero eigenvalue in that direction: **a massless particle**.

Step 3 — count. The vectors $\{v^\alpha\}$, $\alpha = 1, \dots, \dim G - \dim H$, are linearly independent (if some combination $c_\alpha X^\alpha \phi_0$ vanished, that combination would be an unbroken generator, contradicting the split). Hence M^2 has at least $\dim G - \dim H$ zero eigenvalues, and

$$\# \text{ Goldstone bosons} = \dim G - \dim H = \dim(G/H).$$

That is Goldstone's theorem. Note that it comes out as a statement of *linear algebra* about the Hessian of the potential — no dynamics, no loop expansion, no quantum mechanics beyond the identification of eigenvalues with masses.

The geometric restatement, which is worth more than the algebra. The vacuum manifold is G/H , of dimension $\dim G - \dim H$. Every direction *tangent* to it is a direction in which the potential is exactly flat — because V takes the same value at every point of the orbit. Flat direction = zero eigenvalue of the Hessian = massless mode. The Goldstone bosons are literally coordinates on the vacuum manifold, and their number is its dimension. This is why Lecture 24 could see the whole thing by looking at a picture of the Mexican hat.

Consistency checks.

- $SO(2)$ fully broken (Lecture 24): $1 - 0 = 1$ Goldstone. ✓
- $SO(2)$ unbroken (Example 1): 0 Goldstones, both fields massive and degenerate. ✓
- $SO(3) \rightarrow SO(2)$, e.g. a triplet with $\phi_0 = (0, 0, v)$: $3 - 1 = 2$ Goldstones and one massive radial mode. The vacuum manifold is the two-sphere $S^2 = SO(3)/SO(2)$.
- Electroweak: $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$ gives $4 - 1 = 3$ would-be Goldstones — which, once the symmetry is *gauged*, become the longitudinal polarizations of W^+ , W^- , Z , leaving one physical scalar, the Higgs.

One caveat the classical argument hides. What has been proved is a statement about the *classical* potential. The theorem is in fact exact — it holds order by order in perturbation theory and non-perturbatively — but proving that requires the current-algebra version: $\langle 0 | j^{\mu A} | \pi^\alpha(p) \rangle = i f_\pi p^\mu \delta^{A\alpha}$ together with $\partial_\mu j^{\mu A} = 0$ forces a pole at $p^2 = 0$. The classical derivation above is the one that shows *why*; the current-algebra one shows that nothing can spoil it.

Summary of Lecture 25 — the logic in ten lines

Added explanation (not in the original notes) — Summary

1. The massive mode of the broken phase is the *Higgs* h ; the massless one is the *Goldstone* π . The same theory has two descriptions, $\mathcal{L}_\theta$ (Cartesian, manifestly renormalizable) and $\mathcal{L}_\pi$ (exponential, shift symmetry manifest), related by $v + \chi + i\theta = (v + h)e^{i\pi/v}$. [p. 23*****]
2. Expanding that relation gives $\chi = h - \pi^2/2v + \dots$, $\theta = \pi + \pi h/v + \dots$: a *non-linear field redefinition*. It changes the Lagrangian and the Green's functions but not the on-shell S -matrix. [p. 24*****]
3. Price paid: $\mathcal{L}_\pi$ contains non-renormalizable operators. Generic fact — any non-linear redefinition of a renormalizable Lagrangian produces them. [p. 25*****]
4. But there are *two kinds*. Genuine ones (Fermi, massive Yang–Mills, gravity) signal an incomplete theory and show up as amplitudes growing with energy. Spurious ones are artefacts of field coordinates. The diagnostic is the amplitude, not the operator dimension. [p. 25*****]
5. Computing $\pi\pi \rightarrow \pi\pi$ in $\mathcal{L}_\pi$ at tree level: only $\frac{1}{v}h(\partial\pi)^2$ contributes, through s -, t - and u -channel Higgs exchange. [p. 26*****]
6. The result $i\mathcal{M} = -\frac{i}{v^2}\left(\frac{s^2}{s-m_h^2} + \frac{t^2}{t-m_h^2} + \frac{u^2}{u-m_h^2}\right)$ tends to $-\frac{i}{v^2}(s+t+u) = 0$ at high energy: the growth cancels, the non-renormalizable operators were harmless. [p. 27*****]
7. Below m_h one can describe the Goldstone alone. The rules are: π is a real scalar with a shift symmetry. This leaves $\frac{1}{2}(\partial\pi)^2$ at renormalizable level and $c[(\partial\pi)^2]^2$, with $[c] = [M]^{-4}$, as the leading interaction. [p. 28*****]–[p. 29*****]
8. In the EFT, $i\mathcal{M} = 2ic(s^2 + t^2 + u^2)$: now the growth does *not* cancel (squares, all same sign). The EFT announces its own cutoff. [p. 30*****]–[p. 31*****]
9. *Matching*: expand the full amplitude for $s, t, u \ll m_h^2$ and compare. This gives $2c = 1/(v^2 m_h^2)$, i.e. $c = \lambda/m_h^4$, and a cutoff $E \sim m_h/\lambda^{1/4}$. Each UV completion predicts a different c , so measuring c , m_h and the Higgs self-couplings tests the model. [p. 32*****]–[p. 33*****]
10. Generalizing to any compact G : split the generators into unbroken ($T^a\phi_0 = 0$, closing a subalgebra $\mathfrak{h}$, giving $G \rightarrow H$) and broken ($X^\alpha\phi_0 \neq 0$, not closing, not forming a normal subgroup). Differentiating the invariance of V and evaluating at ϕ_0 gives $M^2(T^A\phi_0) = 0$ — one massless mode per broken generator, $\dim G - \dim H$ in all. [p. 34*****]–[p. 42*****]

Added explanation (not in the original notes) — Where this is going

Two threads are now fully in place and about to be tied together.

From Lecture 23: a gauge boson cannot simply be given a mass. The longitudinal polarization becomes physical, nothing controls its interactions, and amplitudes grow like $g^2 E^2/M^2$.

From Lectures 24–25: spontaneously breaking a *global* symmetry produces exactly $\dim G - \dim H$ massless scalars, which interact only derivatively and are coordinates on the vacuum manifold.

Make the symmetry *local*. The Goldstone fields are then precisely the parameters of the gauge transformation — one can gauge them away entirely (“unitary gauge”), and the degree of freedom does not vanish: it reappears as the longitudinal polarization of the gauge boson,

which thereby acquires a mass $\propto gv$. The counting is exact, one would-be Goldstone per massive gauge boson. What is left over is the radial mode: the Higgs boson, whose exchange is exactly what cancelled the bad energy growth in [p. 27*****] and will do the same job for longitudinal gauge boson scattering. That is the Higgs mechanism, and this lecture has supplied every one of its ingredients.

Note also that the matching exercise of [p. 31*****]–[p. 33*****] is not a side remark: it is the template for how the Standard Model is actually tested. Measure the low-energy coefficients, find the predicted scalar, check that its mass and self-couplings agree with the coefficients. The Higgs was found at 125 GeV; the self-coupling measurement is still in progress.